



Just energy transitions to low carbon economies: A review of the concept and its effects on labour and income

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ABSTRACT

The global environmental and social crisis has motivated the renaissance of just transitions, this time centred around the effects on labour and income distribution of the shift to a sustainable energy mix. This renewed conception has gone beyond its origins in the US unions in the 1970s to become a key element in many discourses worldwide, with different nuances. Over the last few years, it has found a place in the SDGs and gathered strong support from the ILO. As political frameworks and literature multiply, this work presents a systematic review of the concept, its evolution and the methodologies devoted to studying it. We make three major points. First, the empirical state-of-the-art has blurred the lines of the methodological classifications found in previous studies: CGE and IAMs are two of the most popular methods, with a growing interest in hybrid approaches and system dynamics. Additionally, while most studies focus on the national level, the regional level is gaining momentum to illustrate the asymmetric effects of the transition. Second, a certain consensus has been detected to point out a positive, yet relatively small, impact of the transition on net employment and a negative effect on income distribution. Third, there is a scarcity of research about income dynamics, while studies on labour are blind to the effects of the transition on the quality of jobs, the evolution of working hours and gender. These gaps have inspired the elaboration of a research agenda that is presented at the end of the work.

1. Introduction

Societies are currently struggling with a twofold global crisis. On the one hand, humanity surpassed such ecological limits as climate change, the biodiversity extinction rate and the nitrogen cycle for human uses a decade ago [1]. On the other hand, poverty, inequality and discrimination are widespread and have been exacerbated by such events as the crisis in 2008. Environmental deterioration and socioeconomic problems are the two main challenges of the century [2].

In this vein, during the past few decades, there has been growing interest in the notion of “socio-ecological transition”, i.e., the change from a model of interaction between societies and natural systems (socio-ecological regime) to another model [3], with the aim of stopping or slowing down the said problems.

The energy concerns linked to the current socio-ecological regime [4], such as the dependence on fossil fuels, the level of energy-related greenhouse gas emissions, the expected increase in energy consumption despite efficiency [5,6] and the peak oil phenomenon [7,8], inter alia, suggest the need to focus on the subsequent energy transition to a low

carbon paradigm.

Not only does such an energy transition imply changes in energy generation, distribution, storage and usage [9,10], but also a complex set of interconnected social, political and economic rearrangements [11]. It is in the convergence of energy transitions and socioeconomic concerns where the concept of a “just energy transition” arises. Even though this idea has considerable grounding in contemporary history, it has experienced a revival and evolution in recent years [12,13] motivated by the said dual crisis [14,15].

Historically, while past energy transitions were involuntary, the present transition is a complex policy-led process with time goals [16–18]. A set of public policies is aimed at stopping the problematic while, at the same time, managing the resulting consequences for natural and human systems [19], since markets cannot be totally trusted [20], especially when it comes to compensating for the negative side effects.

As this concept gains momentum, this article proposes a systematic review of just energy transitions to low carbon scenarios, high on renewable sources, and their implications for labour and income, as well

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as on the methodologies used to analyse them.

The next section sets the goals of the revision and breaks down the review process. The third section is devoted to delimiting the concept, its evolution and principles, with special attention given to the ILO framework. The fourth summarises the methodologies used by researchers to study the just energy transition; while the fifth gathers the most relevant results for labour, income distribution and the main institutional and financial barriers. Finally, a proposal for a research agenda is elaborated in the sixth section and the conclusions of this work are set out in the seventh.

2. Research goals and review methodology

The goals of this review are:

First, to clarify how the concept of “just energy transition” emerged, evolved and gained the attention of multiple and diverse agents.

Second, to determine how the concept has been studied by researchers addressing the effects on labour and income, the methods that have been employed and the results reached. Subsequently, the analysis must determine to what extent the empirical results are conditioned by the applied methodologies, and whether the studies are consistent or not with each other.

Third, to assess these efforts and detect their strengths and weaknesses in order to propose a research agenda accordingly.

The method used to elaborate this paper consists of an unweighted systematic review [21] structured in the following phases (Fig. 1):

Phase 1. Contextualisation. A revision of reports and working papers from international organisations is done, as well as a review of the most commonly referenced articles in related academic projects. This leads us to conclude that a review on just energy transitions must focus on labour and income.

Phase 2. Searching in scientific databases and screening the results. Diverse search strategies are launched in order to retrieve papers containing the different conceptual combination detected in the previous phase. Once the total amount of articles found in such databases has been gathered, they are classified to determine their interest to this research by relying on topicality and conciseness. The theoretical-conceptual revision includes the most relevant literature, while the effects on labour and income especially covers the case of developed countries, due to the relative scarcity of studies based on data from less developed

areas.

Phase 3. Broadening references. From the total amount of selected papers, a search for more references is done by checking the bibliography their authors used. The recovered references are processed as done in the previous phases.

Phase 4. Synthesis. The synthesis containing the main results of the review is presented in the following sections.

This methodology shares the same limitations as other systematic reviews in the field of energy research, as detected by Sovacool et al. [21], i.e., it is a research strategy intensive in resources, narrowly focused on restrictive questions, with an implicit preference for quantitative research and led by an additive approach. With the aim of avoiding these inherent limitations, we have taken the precaution of including a conceptual and institutional overview of a qualitative nature in the first sections to compensate for the possible quantitative bias and the merely additive narratives derived from the review of methodologies.

3. Concept, evolution and principles

The concept of “just energy transition”, in the spotlight of this review due to the energy concerns associated with the current twofold crisis, as stated in the introductory section, is a part of broader concepts that we have to address first (Fig. 2). Three conceptual layers can be distinguished. In a first layer, there is the idea of a socio-ecological transition. A second layer would be the concept of energy transition. In a third and last layer, we can find the notion of just energy transition, which arises as one of the diverse possibilities of shaping an energy transition.

In the first layer, the transitions to a sustainable regime are a broad category that alludes to a process of shift from one socio-ecological regime to a more sustainable regime [1,4,22–26].

In the second layer, energy transitions, as one of the multiple dimensions of a socio-ecological transition to sustainability [27–31], imply a process of shift that affects the generation, distribution, storage and use of energy [9,10] and causes rearrangements in policies, economies and societies [11]. There are diverse formal definitions that consider different elements in energy transitions. Sovacool [31] gathers the most relevant ones so that the distinguishing elements can be seen. One the one hand, some notions of “energy transition” focus on energy sources and technologies, sometimes connoting a mere change in fuels and in the technologies used to exploit them [32,33], specifying that the transition refers to an economic system [34], an energy system [35], or to the timing from the introduction of a new source to the moment when it reaches a notable market share [36]. On the other hand, energy transitions are more frequently pictured as complex phenomena made up of a set of changes affecting resources, services and agents [37]. These phenomena are defined as co-evolution processes referenced to socio-technical systems, or regimes in general [38], implying changes in their tangible elements (such as technologies, infrastructures and markets), in their agents and networks and in the foundation of their own socio-technical system [30,39,40]. In any case, the essence of the transition, as understood today, is not the shift in the energy mix, but the attempt to prevent the escalation of tropospheric temperature. Accordingly, it implies the decarbonisation of the energy supply on a global scale [41].

In the third layer, the concept of “just energy transition” adds the concerns of justice and equity to the previous one [13,42], even if little attention has traditionally been paid to energy justice [43] in comparison with energy sustainability.

The evolution of the concept has been as follows (Fig. 3):

The roots of the current notion for “just energy transition” are in the union movements of the North American energy sector in the 1970s. The unionist Tony Mazzochi was, to the best of our knowledge, the first contributor to shape the concept of “just transition” on behalf of the struggle of the Oil, Chemical and Atomic Workers International Union

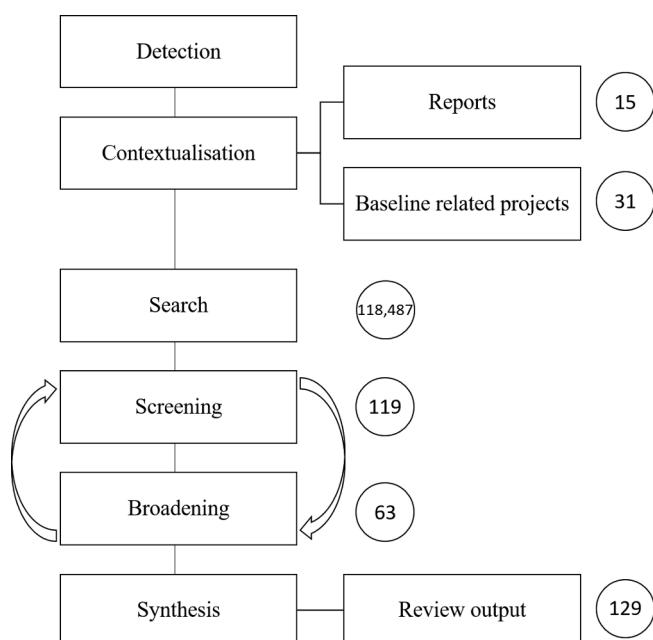


Fig. 1. Review process and reduction of references Source: Own elaboration.

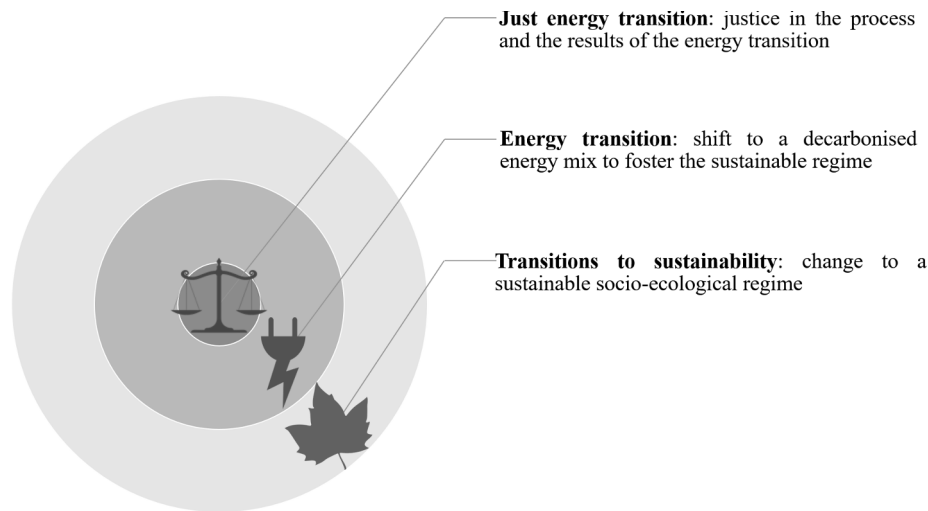


Fig. 2. Conceptualisation of a just energy transition Source: Own elaboration.

(OCAW) to grant propitious labour conditions to the employees of several refineries [12].

The original concept served to unify the social environmentalism of the 1950s and elements of occupational health and safety that influenced the environmentalist claims in the next two decades, closely linked to climate and environmental justice [44].

The diffusion of the concept occurred in the 1990s, when the notion was adapted to other unionist branches and a Just Transition Alliance was set up. In the last two decades, while the use of the concept has declined in the American context, the idea has been considered enriching abroad; in fact, it has been significantly present in the United Nations and the European Union.

Over four decades, it has transcended its original scope to become a global idea that has been included in the mainstream debate, specially linked to the energy transition. The main efforts to study and analyse the concept are currently being carried out in the framework of the International Labour Organisation, which has sponsored it in the Sustainable Development Goals [2,45].

Conceptually, the call for a just energy transition is connected to the concept of “energy justice”, which is disaggregated into distributional, procedural and recognition justice [46–50]. Distributional justice refers to the way in which societies share the positive and negative effects of energy policies; procedural justice stresses the importance of the just

participation of all stakeholders in the process of policy development; while recognition justice involves the delimitation of adversely affected collectives. Therefore, although justice in energy transitions is frequently understood as a final result, justice must also be seen as a process [51].

Due to its wide diffusion, the concept has been analysed through different approaches, of which four can be emphasised, as observed by the Just Transition Research Collaborative [12]: 1) the “statu quo approaches”, which consider a voluntary, private and market-based greening of the system that maintains its capitalist traits; 2) the “managerial reform approaches”, in which the public sector assumes a higher importance, without changing the inherent capitalist balance of power; 3) the “structural reform approaches”, which highlight the need for a redistribution of power and the existence of social property for key assets and activities; and finally 4), the “transformative approaches”, which support a radical change of the economic system, as it is assumed to be responsible for all the negative socio-economic and environmental outputs. International organisations tend to position themselves between the managerial and the structural approaches, while unions and non-governmental organisations are closer to the transformative approach.

These approaches involve two different views regarding economic growth. On the one hand, some authors consider that economic growth

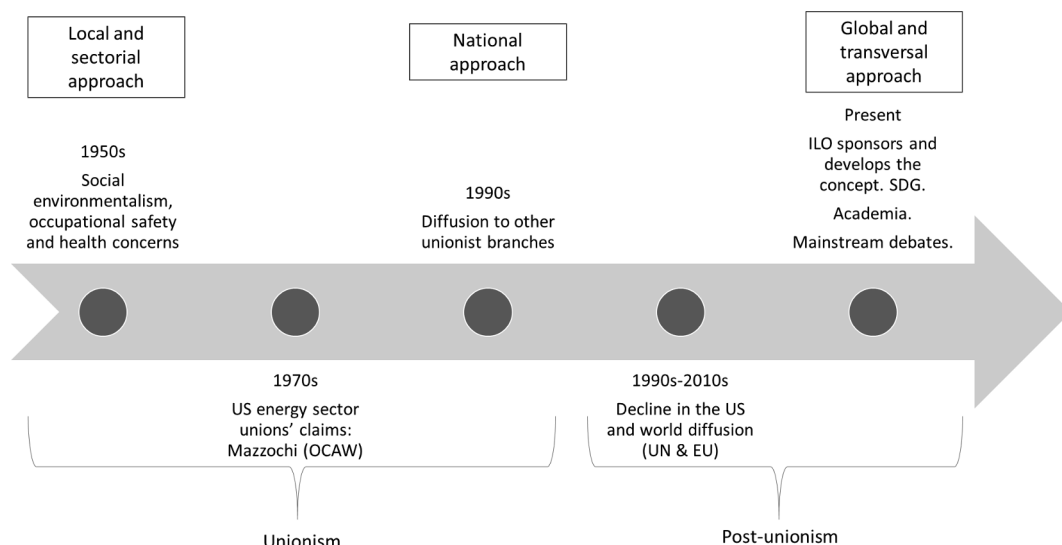


Fig. 3. Just energy transition evolution timeline Source: Own elaboration.

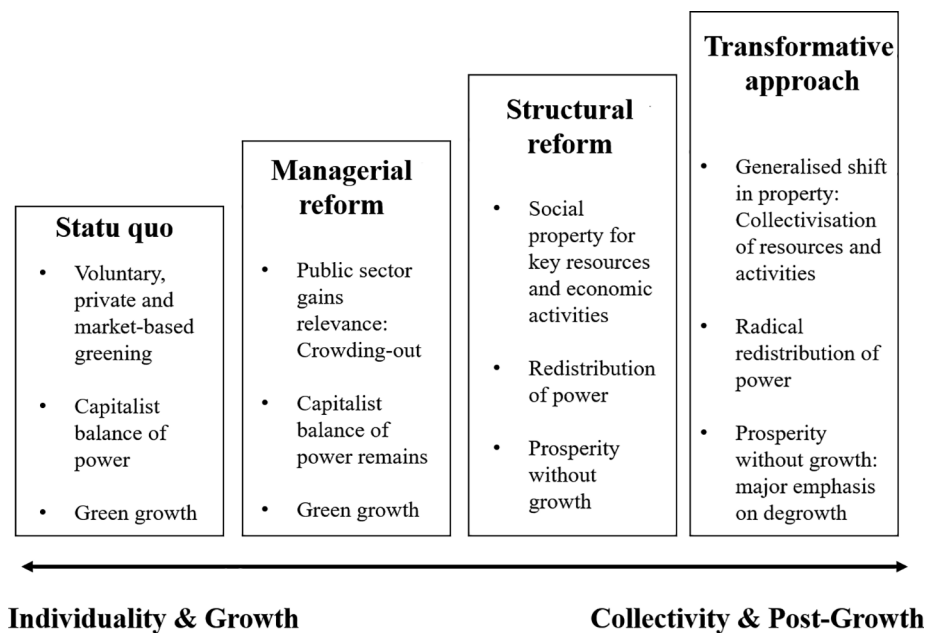


Fig. 4. Approaches to the concept of just energy transition Source: Own elaboration.

is a desirable output of the transition because: 1) the cost of decreasing the emissions responsible for climate change is insignificant in comparison with the cost that an economic system will have to face if climate change worsens [52,53]; and 2) the cost of decreasing the emissions is lower than the benefits of the transition towards a green paradigm [54]. On the other hand, other authors consider that the transition requires a dramatic change in our societies, away from the growth paradigm, and instead supporting a “post-growth perspective” or a “prosperity without growth” approach, which implies a controlled degrowth in the use of energy and materials [55–59] (see how both dimensions are related in Fig. 4).

To illustrate these approaches, we have revised the discourse and the theoretical developments of the International Labour Organisation, the main contributor and sponsor of the concept nowadays. The ILO [2] considers that the transition offers three opportunities and three challenges (Table 1).

The trigger of the transition is the joint effect of demand and investment shifts towards green sectors and the subsequent displacement of resources from non-green activities, mainly guided and propitiated

by public policies. These changes will cause a parallel creation and destruction of direct jobs and propagate through supply chains and income variations affecting the whole economy, therefore creating and destroying indirect and induced jobs. These effects have also been observed by other organisations [60,61]. The net balance of jobs would be a result of demand levels, investment, trade flows and labour elasticity. The difficulties in adjustment could come from unexpected job losses, the age and geographical identification of employees, the absence of nearby alternatives and the shortcoming of political programmes. It must be emphasised that, from this perspective, not only must gross and net effects on employment be considered, but also job quality, employee skills and gender equality, a key worry for the ILO [45].

Concerning income, more ecological goods and services could lead to higher prices, and thus to a smaller amount of available income to spend on other goods and services. Inversely, more profitable investments and efficiency could boost demand in subsequent time periods, through a shift from energy demand, with low elasticity, to non-energetic goods and services with a higher elasticity [2]. The outcome in resource consumption and GDP is a priori undetermined (see the

Table 1
Opportunities and challenges of a just energy transition in the ILO framework.

Opportunities	Challenges
1. Creation of net employment: -Double dividend -Economic growth via investment 2. Improvement of labour quality: -Higher wages via higher competences and specialisation -Permanent and full-time jobs -Higher labour protection and better working conditions -Highlighted sectors: Agriculture: absorption of new workers in better conditions Recycling: jobs regularisation, female labour, higher intensity and wages Construction: regularisation, competences and health safety 3. Social inclusion: -Reduction of poverty and access to affordable non-pollutant energy: Gender equality -Enhancement of local communities via environmental payments	1. Socioeconomic restructuring: -Ecologisation: Increasing resources productivity -Higher automatization -Variation in the sectorial composition of employment -Implication of employees and social stakeholders -Small labour mobility -Difficulties for small businesses 2. Adaptation to climate change: -Extreme weather conditions and their multidimensional consequences: Poor harvests Scarcity of drinking water Migrations 3. Compensation of regressive political effects: -Higher energy prices and negative effects in income distribution

Source: Own elaboration.

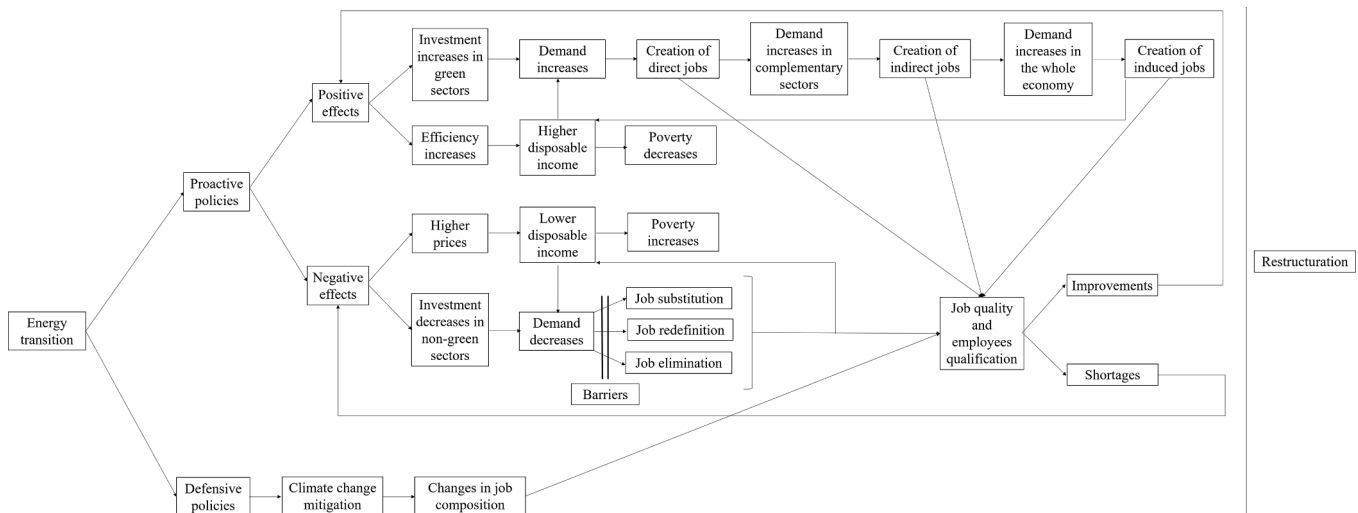


Fig. 5. Theoretical effects of energy transition on labour and income from the ILO perspective Source: Own elaboration.

antagonist forces operating in Fig. 5).

This combination in the ILO framework of a strong public sector with a positive narrative about economic growth, but with some nuances of post-growth, can be identified as a managerial reform close to some elements of the structural reformative approach.

As a concluding point, we propose a definition by combining the previous partial definitions and the elements revised in this section. A “just energy transition towards a low carbon economy” can be defined as a long-term technological and socio-economic process of structural shift [11,37,62] that affects the generation, distribution, storage and use of energy [9,32,33] and causes rearrangements at micro (innovation), meso (social networks, rules and technical elements) and macro (exogenous environment) levels [30,38–40,63], while also ensuring that the desired socioeconomic functions can be accomplished through decarbonised and renewable means of energy production and consumption [41,64], safeguarding social justice, equity and welfare. From a practical perspective, the transition is a public matter of instrument choice (to foster the substitution of energy sources) and instrument change (to adapt the policy scheme in coherence with the multi-dimensional challenges that must be faced) [65].

Scholars have not agreed upon the principles of how to analyse a just energy transition, apart from the need to contrast the net effects of all alternatives [66]. Notwithstanding, along this review, a group of demands and features for a just energy transition have been found. They can be called the “must-have” principles of a just energy transition:

- The transition must be flexible enough to cope with uncertainty and social complexity [64].
- The costs derived from the transition are not the only matter to contemplate. The collectives that have barriers to access the benefits of the transition, generally low-income households [67,68], must have their barriers removed and benefit in equal conditions.
- Policies must consider and respect the rights of local communities and solve the historical injustices caused by the diversity of perspectives among stakeholders. Scholars propose community-based participatory decision-making and increasing cross-border activism supported by research [47].
- The transition policies must ensure qualified jobs and human resources, enable retraining, focus on education, health and creativity, provide certainty to industries in transition and facilitate new businesses [69].
- Policies must be consistent, driven by a long-term vision and fuelled by cooperation between stakeholders, with special emphasis on the participation of workers [42].

4. Methodologies to analyse the just energy transitions

The predominant method to analyse the topic under study is the elaboration and application of models. The modelling of just energy transitions emerged from the attempt to integrate a growing environmental awareness with the Keynesian and Post-Keynesian approaches. This branch has started to develop models to assess economic inequality and job markets in connection with environmental limitations and alternative economic paradigms [70], contributing to filling a gap in the research into transitions [71].

As the models are diverse, they are classified into groups and sub-groups [70,72]. The most widespread classification is also the most categorical. It considers the existence of numerical and analytical models according to their base; while earth system models, Integrated Assessment Models (IAMs) and Computable General Equilibrium models (CGE) are considered according to their level of detail [73].

Numerical models usually apply Input-Output (IO) analysis. Input-Output methods are recurring in the literature [74] and consist of a set of tables that show the flows of goods and services between intermediate units and final units [15]. They are useful as transformations propagated through supply chains, but they also present methodological limitations [15,75]: the difficulty of considering time lags, measuring feedbacks between prices and quantities, the linearity and invariance of the coefficients, the generalisations of sectors, the homogeneity of outputs, and the lack of economies of scale. What is more, their reliability is inherited from primary data sources and they tend to overestimate job creation. Input-Output tables are also scaled on a national level, so it is difficult to use them in a regional or local perspective [72]. Analytical methods are commonly used for regional or local studies instead. They are more transparent, but cannot predict indirect or induced jobs.

Regarding the perspective of the modelling, there are two main, widely used approaches: the top-down approach and the bottom-up approach. When applied separately, these approaches have significant demonstrated weaknesses [76,77]. The top-down approach relies on a static Input-Output methodology that cannot offer an accurate level of detail in sectorial estimates. Moreover, it is not appropriate to capture the shifts caused by investment variables and offers other inconveniences attached to the timing of the model. The bottom-up approach configures limited analytical models dependant on accounting techniques that do not provide the big picture. It would therefore be useful to combine both approaches; benefitting from their strengths and avoiding the weaknesses. This combined approach is called “hybrid” and has been increasingly applied in recent research [78].

Apart from this classification, and according to the level of detail

[73], earth system models contemplate a simple economic framework and a complex environmental background; while computable general equilibrium models rely on a complex economic frame despite a simpler environmental background. IAMs balance the complexity of environmental and economic systems. In methodological terms, increasing attention has been paid to the use of IAMs, such as IMAGE [73], ReMIND-R [79] and EPPA [80], to the point of becoming the primary method for studying the transition.

While this classification gathers the most common terminology, it should be used with caution and an open perspective, since the empirical state-of-the-art is surpassing such restrictive classifications, as the existence of methodological intersections and hybrid approaches proves.

Apart from this caution, we have found, along the review, three increasingly generalised claims to remark. First, the growing call to give Social Sciences an opportunity to enrich this discussion [81,82], besides the encouragement of the use of case studies [81,83], the application of systemic thinking [47] and the cooperation between fields [48] through transdisciplinary analysis [84]. Second, the increasing interest in socio-metabolic approaches [4,85,86] to understand the paths of societies and provide thorough comparisons, reinforcing the use of case studies [87]. Third, the explicit argumentations in favour of demand approaches, based on the risk of limiting the capacity of the models when it comes to the caption of social problems and impacts on welfare with full employment assumptions [88]. Concurrently, most of the published works are still based on growth paradigms. A lack of studies focusing on structural and transformative approaches had already been noticed long ago [38], even though it is a recurring alternative found in theoretical studies, as stated in the conceptualisation.

Once this methodological framework has been set up, we are in a position to delve into the empirical effects set out by works in the field, as described in the following section.

5. Empirical effects

5.1. Effects on employment

Most studies estimate the effects of the energy transition by calculating the variation in the number of direct, indirect and/or induced jobs [22]. According to their geographical context, we have gathered them into three groups: supranational, national and regional studies. Regarding studies at the European supranational level, the following can be highlighted:

Cambridge Econometrics [22] used the model E3ME, considering the EU 2020 objectives, and found that this strategy would lead to a small creation of jobs via an increase in the investment variable. However, it was unable to offer a detailed disaggregation of job flows between sectors.

The Green Jobs study, by Cambridge Econometrics, also focused on the 2020 horizon, but data were taken at a moment of economic recession and did not consider major changes in the labour market or in technologies. The study concluded that the variation in the number of jobs in Europe would be small and the most significant problems would be the consequences of a negative impact of labour mobility on aged workers and skill shortages in certain sectors. The role of labour mobility is a key to determining whether the effects are positive or negative.

Focusing on the 2030 horizon, both for the EU and other areas, E3ME has been updated and linked with the Warwick Labour Market Extension model [89]. The results reinforce those obtained in the previous application of E3ME [22]: again a positive outcome is found in employment due to investment, especially in energy, construction and related materials sectors. Cambridge Econometrics [90] additionally provides the impacts of a shift to low-carbon transport following the energy transition through FEF (E3ME model) and ELAB: their conclusion was that a potentiation of hybrid vehicles would have beneficial

results, while a proliferation of battery electric vehicles would destroy jobs.

NEMESIS and ASTRA models have also been applied to the 2030 horizon [91]. NEMESIS projected more optimistic conclusions, highlighting the benefits of an accelerated deployment scenario. ASTRA was more pessimistic due to the differences attached to the investment variable.

At this moment, the focus is on 2030, but especially on 2050, according to action plans [18,92], or even on 2100, to deepen the research ambition [93]. Nevertheless, the methodological approaches and the results of the cited studies are revealing and must be considered.

Towards the 2050 horizon, the E3ME and the GEM-E3 models have been used [22], as well as the LUT model [94]. The E3ME estimates a net increase in employment equal to 1.2%, due to the fall in oil prices caused by the decrease in fossil fuel supplies, the increase in investments, the use of public revenues, the loss of jobs in polluting and intensive energy sectors, and the rise in electricity prices. The GEM-E3 also concludes that a net positive, yet small, impact is likely to be expected, despite the scenario. Notwithstanding, skill shortages push wages up and negatively affect the level of employment. In contrast, LUT predicts a high positive result, since renewables would contribute to 80% of employment by 2050 and the transition would potentiate economic growth, productivity and efficiency.

Concerning the GEM-E3 model, Fragkos & Paroussos [75] follow a hybrid approach, relying on a more remarkable Neoclassical background, thus prioritising supply rather than demand. As a result, they estimate creation and reallocation of jobs of around 1% of the European labour force. Again, a positive yet insignificant outcome is found. Skill shortages are not so important in this update, probably because of the methodological background.

Additional studies in the EU conclude that the transition has an especially positive effect on the energy sector. However, the measures implemented cause a rise in prices that negatively impact production and employment in the rest of the sectors. According to their authors, the negative effects can be partially tackled through market integrations [95].

In contrast with the ex-ante nature of these works, studies that look at the past through counterfactual methods can also be found. Markandya et al. [15] uses the Input-Output method applied to past data on electricity and gas from a multiregional perspective to find spill-over effects caused by changes in the energy system. Its conclusion is coherent with the preceding ex-ante studies: an increase in employment can be detected after the implementation of sustainable energy measures.

Besides the European supranational level, there are also abundant studies applied to the national and regional contexts that most often emphasise the uneven distribution of impacts across them [42]. Among these studies, certain cases are presented as paradigmatic: Germany is the most analysed and cited case [96] under a diverse set of methods referenced below; the Netherlands, using the CGEM ThreeME model [97]; Italy, with a Lotka-Volterra growth model [98]; France, using directly the Input-Output methodology [99] and the EUROGREEN model [100]; and Catalonia, using the MuSIASEM model [86].

Germany has generated more literature than any other country, in most cases to point out the negative effects of the German energy strategy, especially on employment and equity [101]. However, these studies are frequently conceived from an exclusive point of view and do not aspire to observing the big picture, since they only focus on electricity (mainly as a reflection of the Energiewende, which is entirely focused on electricity) [102].

Notwithstanding, the German case has served to clarify some transcendent conclusions. Fischer et al. [101] offer two important facts. First, the key to analysing the job variations caused by the transition is not gross employment, but net employment: gross employment shows a positive trend as long as renewables create new job positions; however, net employment measures whether the destruction of jobs on older,

conventional sources are replaced or not by the creation of jobs in renewables. Second, the macroeconomic effects of the transition are deeply determined by the speed of its implementation. The speed is a political variable that can be used to influence the outcome, determining the level of macroeconomic and environmental costs assumed by society. Many studies support a fast transition based on R&D, the backing of social agents and the development of cost-competitive renewable technologies [103], apart from the context of globalisation [104]. In contrast, other authors are sceptical when it comes to speeding up the process [105], or they are even sceptical about the full concept [65]. Apart from the speed, Fragkos & Paroussos [75] add that the outcome depends on the situation of the economy regarding the production possibilities frontier, the prices of primary factors, the conditions of flexibility in the job market and the labour intensity.

Focusing on solar and wind energy, the two strategic sources for the transition; while most employment is created by investment on installations, a growing number of jobs are generated in maintenance and operation services. While the importance of investment is clear, export markets are also a key variable, in line with other studies that complain about recurrently omitting it [15]. Markandya et al. [15] find, through their counterfactual method, that employment creation follows geographic patterns and is more dispersed than expected. In this vein, and looking to future estimations, Sievers et al. [69] conclude an overall positive impact on growth and employment, but the worrying nuances are also found in the geographical distribution of impacts across Germany. The construction sector is again the winner in terms of employment along with the energy sector, while the rest of the economy suffers the negative impact of the rise in energy prices.

Likewise, Blazejczak et al. [106] find a positive net effect on economic growth, boosted by investment and dependent on market flexibility. Effects on employment are small and positive if the labour market is rigid, and the results tend to improve as it gains flexibility. The shift in energy generates a restructuring process that requires an improvement in skills.

The PANTA RHEI model, applied to Germany, concluded that the deployment of renewable energies would increase net employment, especially from 2020 onwards [74], pointing out the relevance of international trade flows in determining the result and concluding that the future effect on gross employment would not be as fast as it was in the past.

Apart from Germany, other recurring national cases in the literature are the Netherlands, Italy and France.

In the Dutch case [97], the model reflects the possibility of creating 50,000 full-time jobs by 2030, linked to a 0.85% increase in the GDP. The effects are derived from the analysis of electricity generation and prove the potential of wind and solar sources, with a higher labour and capital intensity. The capital intensity is the factor that, once again, serves to accentuate the importance of the investment variables. The authors conclude that renewable energy systems are more beneficial for the Dutch economy than the conventional ones.

In the Italian case [98], a Lotka-Volterra growth model underlines the link between employment and income distribution. The introduction of system dynamics provides flexibility and allows the feedbacks occurring among the modules of the model to be visualised. A certain degree of dynamisation is also looked for through simulation and prediction scenarios. The conclusion on jobs is consistent with the previous studies: the model predicts a 1.2 million increase in the number of jobs, showing again the relevance of the investment variable. However, this increase adds pressure on wages and the GDP falls slightly via investment and capital accumulation. The final situation is strongly determined by wage elasticity in the energy sector, on which the analysis focuses.

In the French case [99], as data is not updated, the study does not consider the new political commitments. Besides, the energy transition is understood as an evolution towards a lesser usage of fossil fuels, focusing on the taxation of CO₂ emissions. The study estimates the jobs

in each economy branch and compares these to the national average by using five variables: final import rate, intermediate import rate, tax and subsidies, the portion of jobs in added value and wage level. The different results between branches are caused mainly by wage level and added value, although there are branches of the economy where the method is inapplicable. The results are coherent with those estimated at a European level. In order to update this case, the EUROGREEN model has been revised [100]. EUROGREEN, built on system dynamics, analyses in a novel way the reaction of the French economy to the application of measures compatible with green growth, social equity and degrowth scenarios. It concludes that green growth serves to reduce greenhouse gases emissions to the detriment of employment; social equity policies coordinates the same environmental performance than green growth with better social outcomes at the expense of a higher budget deficit; and finally, degrowth reaches the best environmental and social results, but also the highest deficit shares even if wealth taxes are inputted.

Continuing with the national level, but far from the European cases, we have found a work devoted to the Global South that analyses just energy transitions in a sample of less developed countries [107]. It observes the profile of these countries through indicators and their degree of current and potential compliance with the SDGs. Additionally, they gather statistics and short-term estimations from national authorities and international organisations regarding the impacts on employment levels. Under these indicators, South Africa, India, rural Vietnam, Philippines, Morocco and Jamaica have experienced, and will experience, the positive impacts on employment of renewable and decentralised means of energy production. This report on the Global South has palliated the absence of cases and reliable data from less developed countries that we have found.

Finally, beyond the national studies, there are pieces of evidence that seriously recommend taking the regional contexts into account [108]. It is necessary to highlight the MuSIASEM model that has been applied to Catalonia [86]. MuSIASEM divides economic activities into general groups and considers just two scenarios. It coherently predicts an increase in jobs in the absence of skill shortages.

After this revision on the side of labour, the following points summarise the main findings:

- Analyses have to observe the effects by countries and regions, as they would be unevenly distributed within them [42].
- Studies have shifted from analysing the consequences of taxes on employment [99], to progressively widening their scope. This is a strongly recommendable practice, since the effects of taxes are only one of the diverse policies that determine the transition.
- Studies are progressively focusing on net employment, which is a strongly recommendable practice, but are only interested in volumes, which is not recommendable. Other important factors, such as the quality of jobs or the number of working hours, are constantly omitted, as noticed long ago [109], even if they are statistically significant variables [110]. Therefore, the creation of poor-quality jobs would be considered a positive outcome of the transition in these studies; while, in fact, it should be considered as a negative outcome, since poor-quality jobs damage the environment and do not improve the social situation [45]. Papers also tend to use and reuse employment factors, even when they apply a wide range of methodologies [111]. The reuse of factors is not a recommendable practice, as it leads to inaccuracy. Likewise, research tends to focus exclusively on the energy sector and obviate the rest [95,101], or estimates the diffusion from the said sector to other branches. Studies observing the big picture are few. This evidence again supports the necessity of applying systemic thinking.
- In the sectorial analysis, the most positive employment impact is recurrently found in construction and the green energy sectors. Overall, a positive, yet small, impact is found on the economy. These results are less favourable than expected, as renewable energies are

Table 2
Classification of the reviewed empirical studies.

Units	Methods	Methodological traits	Results	Time span	References
World regions	IAM	A human and an earth system are interlinked through land and emissions. It is triggered by such drivers as population, economy, policies, technology, lifestyle and resources to calculate impacts on the environment and development. Limitations: inaccurate monetary feedbacks, calibration distortions in the period 2010–2020, lack of specificity for local and national scales, impossibility of testing specific policies and absence of governance system.	As labour supply and income disparity via the Gini index are drivers of the model, relying on the definition of scenarios, the interest of this model is the joint consideration of socioeconomic and ecological dimensions in an integrated approach.	1970–2050/ 2100	Stehfest et al. [73]
	IAM	It hard-links a macroeconomic (Ramsey optimal growth with equilibrium constraints), an energy and a climate module. Capital, labour and energy are inputs to the model. It includes international trade and investment dynamics. Limitations: lack of detail in processes happening inside world regions, renewable supply intermittence is not detailed, difficulty in capturing efficiency potentials, missing technological spill-overs and stocks of knowledge, lack of constraints in bio-energy.	Since labour is an input, the relevance of the model is the parallel consideration of energy-economy-climate relations in an integrated model with trade and investment dynamics.	2005–2100	Luderer et al. [79]
	Employment factors	The total employment is calculated as the addition of key jobs during the transition. Energy system transition model: simulation applied to cost optimality in five-year periods. Innovation: inclusion of energy storage effects.	Considerable positive effects: renewables contribute to 80% of jobs by 2050, and widely exceed the rate of lost jobs. Higher efficiency, productivity and economic growth. Different impacts across regions, but with a general growth potential.	2015–2050	Ram, Aghahosseini, & Breyer [94]
	Methodological combinations	As it is a wide-picture report, it combines different methodologies for each topic under focus. It is the only reviewed piece that provides an advanced understanding of gender impacts and also an estimation of the effects of climate change combined with the impacts of the energy transition. The relation between decoupling and employment is studied through econometric logarithmic regressions. The resource intensity of employment is calculated by distributing resource consumption equally among all registered employment. The impact of human-induced disasters on labour is estimated by adapting Noy's benchmarking index. The effects of heat stress on workers are calculated by applying grid cell data to population cohorts and employment distribution by sector. The employment balance by sectors, wages, emissions, skills and gender is estimated with multiregional IO tables.	Net positive effects: declining energy activities will lose 6 M jobs, while greening energy activities will generate 24 M worldwide. Gender inequality is exacerbated because new jobs are created in industries with higher male-related employment: construction, energy and manufacturing. Climate change-related natural disasters caused the loss of 0.8% working-time globally between 2010 and 2015. Heat stress will suppose a reduction of 2% in working hours globally by 2030.	1995–2030	ILO [45]
World countries	IRENA database indicators	Analysis of IRENAS' database historical indicators on renewable energy-related employment by volume, technology, product, gender and country. Thus, focused on direct observed records for the energy sector solely.	In 2018, 11 M jobs were registered in renewable energy (7% interannual increase), with approximately 33% of them in solar photovoltaic power, 19% in hydro, 19% in liquid biofuels and 11% in wind power. 39% of global employment is in China, followed by the EU with an 11% share. By gender, 32% of jobs in renewables are carried out by women.	2018	IRENA [123]
Global South country sample	Indicators	Analysis of statistics and estimations from national governments and international organisations. The goal is to study the compliance with SDGs and the potentiality of a just energy transition in a sample of Global South countries.	South African government estimates 300,000 new jobs in renewables in 2020, but lacks a net perspective. India created 400,000 jobs in 2015 and could create 1 M by 2022 relying on wind and solar sources. Rural Vietnam is thought to benefit from decentralised energy technologies (no data provided). Philippines reached 9700 jobs in bioenergy and 100,000 in the construction of renewable infrastructure in 2015. Morocco has a greater exposure to the consequences of climate change. The ratio of employment in renewable vs. conventional plants is 5:1 (no overall data provided). Jamaica created 525 jobs between 2015 and 2017 and could reach 40,000 jobs if performs a complete transition to renewables.	Indicators until 2017 and projections to 2020/2030/2050	Hirsch et al. [107]

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Table 2 (continued)

Units	Methods	Methodological traits	Results	Time span	References
US	IAM	Comparison between models. GCAM is a simulation model of partial equilibrium focused on energy and land use that runs in five-year periods. MERGE is an inter-temporal optimisation model of general equilibrium based around economic growth and investment, whose time step is a ten-year period. EPPA is a simulation model of general equilibrium centred on the nexus between energy and economy with a five-year step.	Interest lies in the comparison of models in order to strengthen the design and adoption of measures by policymakers. Main insight derived from the comparison: carbon intensities are the source of disparity in comparisons between models. Lack of clear evidence in labour and distributional matters.	1990–2100	Wilkerson et al. [80]
EU	Regressions	Logistic and Poisson regressions with fixed effects and temporal lags applied to sociodemographic and economic variables regarding the effects of wind energy.	Wind energy has not caused significant injustices in income or ethnicity, but on younger people, with lower levels of qualifications and labour force participation in rural areas.	2008–2017	Mueller & Brooks [122]
	Econometric model	Parameters are determined by statistical techniques applied on time series. Keynesian background. Demand assumes a higher relevance. Electricity prices are used to generate industrial prices and thus consumption. Employment is a result of output, which is pictured by observing consumption, international trade, efficiency, fuel consumption and investment. Income is derived from employment and contrasted with consumption. It contemplates the possibility of non-optimality and imbalances in markets.	The application of E3ME to the EU context has generated considerably negative impacts of –1.6% in employment and –3.4% in GDP.	2009/2050, 2011/2020, 2009/2050	Lewney et al. [89]
		Investment is not totally linked to savings since a stock of capital is available. Only detailed at a European level; the rest of the world is assumed in the 2013 edition (2019 update also includes areas like the US). Limitation: not all roadmap technologies were considered.	Small creation of jobs via increasing investment. Biggest positive impact on construction and energy efficiency materials sectors. Jobs will change scope without major changes in skill distribution. Marginal overall changes in jobs, labour rigidity and skill shortages. Labour mobility as key factor to determine the net effects. Employment net increase of 1.2% due to the fall in oil prices, the increase in investment, the use of public revenues, the loss of jobs in polluting and intensive sectors, and the rise in electricity prices. In the 2019 update, the result is coherent with previous results: an increase of 0.5% is expected in employment and 1.1% in GDP. Investment is again the main cause of this positive result.		Cambridge Econometrics [22], Lewney et al. [89]
	Econometric model	Macro-sectoral model based on CES production functions. Neoknesian effects have a greater influence. Core model plus four modules. Energy cost affects exports, and thus consumption via competitiveness.	Optimistic consequences: Benefits of an accelerated deployment scenario: 410,000 new jobs and an associated increase of GDP equal to 0.24% in the European Union resulting from the total achievement of the 20% RES goal. More ambitious policies could grant a 0.4% increase of the GDP and up to 545,000 jobs in 2030. Pessimistic consequences due to the investment variable and the enhanced impacts of energy cost compensations on labour intensive sectors: energy costs are compensated for through shifts in households' marginal consumption, subsequently reinforcing the demand for less labour-intensive goods.	1990–2030	Ragwitz et al. [91]
	IAM	Built on system dynamics. Neoclassical background complemented by demand-side structural change. System dynamics: Nine modules and a connection between demand and supply. Foreign trade considered and covered. Energy cost is compensated through shifts in marginal consumption that avoids sectors with higher labour intensities.			
	CGE	Hybrid model: parameters coming from related literature. Neoclassical background: focus on supply and optimality leading to market clearing via prices. Investment is closely attached to savings. Alternative options for users, who can choose some variables.	A small net positive impact is expected despite the scenario. Wage problematic. Results more pessimistic than in E3ME model. Substitution and reallocation of jobs constituting around 1% of total labour force. Final effects depend on the PPF, the prices of primary factors, labour flexibility and intensity. Skill shortages are less important.	2009–2050, 2015–2050	Cambridge Econometrics [22], Fragkos & Parousos [75]
	Indicators	Focused on the energy sector. Contrasts the EU North with the EU South.	Positive effect on energy sector employment found in the literature and up to 1% increase in GDP, favourable to Southern Europe. Rise in prices that affect production and employment in remaining sectors. Interconnectivity is highly recommendable.	–2050	Creutzig et al. [95]
IO		Multiregional to find spill-over effects. No scenarios. Counterfactual method.	Increase in employment after the implementation of sustainability measures (530,000 net jobs, 0.24% in 2009). Exports as key factor.	1995–2009	Markandya et al. [15]

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Table 2 (continued)

Units	Methods	Methodological traits	Results	Time span	References
Germany	Review, work-shops & surveys	Qualitative methodology focused on net employment. The macroeconomic effects are determined by the speed of implementation.	Negative effects of the German energy strategy, especially on employment and equity. Negative impacts on employment and higher electricity prices affecting equity receive more attention from the industry, unions and governments than from the general population.	2009–2016 and instant data from 2016	Fischer et al. [101]
	Cost-benefit analysis	Cost-benefit analysis and cost assessment. Solar and wind contrasted. Qualitative, reliant on market argumentation. Focused on electricity.	Most employment is created from investment in installations, but a growing number of jobs are generated in maintenance and services. Employment creation follows geographical patterns and is disperse. Export markets as key variable. The distributive effects are unfair from a social perspective (prices have doubled, affecting low income households, and public subsidies for renewables tend to benefit wealthy citizens). Inequality has been increased by network changes, causing a loss of welfare. Regressive effects could lead to social conflict. Failures in the design of the German transition drive some authors to propose greater trust in market mechanisms, although it is not a generalised claim and most of them propose better policies. Those who believe in market mechanisms perceive that previous studies show a dramatisation of facts and price rises are explained because of beneficial quality increases.	2000–2013, 1998–2013, 1990/2010–2030, 2010 vs 2016, 2000–2015, 1998–2015	Andor et al. [120], Frondel et al. [113], Gawel et al. [119], Heindl et al. [116], Pegels & Lütkenhorst [96], Schlesiowsky & Winter [117]
	Economic metrics	Econometric analysis of massive surveys.	Perceived distributional effects are key, as are the real effects. Support for the statement which says that those who pollute the most, pay the most. Real burdens are not as high as perceived and are manipulated by elites.	Instant data from 2015	Groh & Ziegler [114]
	Economic impact assessment	Economic activities connected to physical consumption and energy are unidirectionally soft-linked for each scenario to a macroeconomic model with regional distributions and federal structures.	Overall positive impact on growth and employment, but a problematic geographical distribution of effects. Negative impact on the rise in prices. Construction sector is the winner.	2010 vs 2030	Sievers et al. [69]
	IO and Economic metrics	Neo-Keynesian background considering nominal rigidities, adjustment delays and dynamic error correction. Top-down approach. It contemplates governmental activities and international investment and trade. The model for Germany has been expanded with the use of IO with the specific aim of calculating sectorial impacts. Employment levels are calculated transitioning from gross output to working hours through prices and productivities.	Positive net effect on economic growth boosted by investment. Effects on employment depend on labour flexibility. The restructuring requires an improvement in qualification.	2000–2030	Blazejczak et al. [106]
	IO and Economic metrics	Energy-economy-climate econometric simulation model based on IO tables. Bottom-up approach. Limited rationality and non-clearing markets. It considers distribution and redistribution of income, as well as financial agents, and acknowledges international dependencies. It runs in one-year periods.	Net employment will increase, especially from 2020. 150,000 net employments can be created by 2030. Import/export flows are key to determining the result: under decreased levels of renewable energy exports, net levels of employment could be negative (i.e., greater job destruction). The future effect on gross employment will not be as fast as in the past.	2000–2030	Lehr et al. [74]
Netherlands	CGE	Neo-Keynesian background. Endogenous capital stock and slow adjustment in prices and quantities. It recognises disequilibrium situations. Country-generic.	Positive impact: 50,000 new jobs in 2030 and 1% increase in GDP.	2010–2030	Bulavskaya & Reynès [97]

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Table 2 (continued)

Units	Methods	Methodological traits	Results	Time span	References
Italy	System dynamics	Lotka-Volterra growth model. Wages and employment are endogenous variables. Long-term horizons are studied based on emissions and GDP variations, while the short-term is analysed in terms of employment and inequalities.	1.2 M increase in the number of jobs, pointing to the relevance of the investment variable. GDP falls slightly due to pressure on wages, investment and capital accumulation. It emphasises the link between employment and income distribution and the advantages of system dynamics. The result is determined by wage flexibility.	1970–2030	Bernardo & D'Alessandro [98]
France	IO	Data not updated and missing important events. Inapplicable for several branches.	Results compatible with those at the European level. The different results between branches are caused by wage levels and the share of labour in added value.	2010	Perrier & Quirion [99]
	IO and system dynamics	Country-specific dynamic macroeconomic simulation model. Combination of Post-Keynesian and Ecological Economics backgrounds. Nexus energy-economy-environment. It models an open economy by assuming the rest of the world. Supply is defined through IO tables with endogenous change and technological progress. There are financial constraints for firms. It contemplates three levels of skills, the variation of wages and working time, as well as the effects of automatization. Inequality is analysed via the Gini index. Households' consumption considers financial variables besides wages and social transfers. It models the public budget balance and recognises the existence of budgetary restrictions. Scenarios: green growth, social equity and degrowth. Limitations: invariance of technical coefficients of non-energy industries, simplified policy framework, lack of specific technologies for RES sources and storage and absence of climate change-economy feedback.	Under a green growth scenario, unemployment increases 3% by 2050, the Gini index increments by 2.5 points in 30 years, GDP grows by 1% and emissions equal to 23% of those registered in 1990. Deficit declines and reaches 1.5% in 2050. Concerning policies for social equality scenario, unemployment falls 2% in 2050, the Gini index declines more than 4.5 points and stabilises by 2040, the variation of the GDP and the reduction of emissions are similar to the green growth scenario. Deficit increases up to 4% in 2050 (3% EU limit exceeded). In degrowth, unemployment falls 7% by 2050, the Gini index drops approximately between 6 and 9 points, GDP falls 0.7% by 2050 and emissions equal to 17.8%. Deficit dramatically hikes to 6.5% in 2050 (EU limit notably exceeded).	2014–2050	D'Alessandro et al. [100]
Poland	Indicators & factors	Case study that cross-checks the national energy balance and the employment attached to it. The estimation is the result of multiplying energy units by employment factors from national statistics and transition plans, some of which are reused (caution).	Despite the high activity in solid fuels, the variation of employment between the BAU and the transition scenarios is small and can be broken down into two trends: a negative impact on jobs until 2030 and a positive impact from 2030 until 2050. The overall increase is 21% in 2050. Jobs in fossil fuel related sectors are totally phased out. The creation of green jobs (approximately 120,000 only in the energy sector by 2050) is higher than the European average due to a higher renewable capacity. A fast deployment of renewables significantly increases the creation of jobs. However, jobs in primary fuels are expected to increase until 2020, representing 22% of the total estimation. The effects of the economic crisis have affected the potential of renewables to create jobs.	2014–2050	Behrens et al. [4]
Sweden					
Spain					
Catalonia (Spain)	Social metabolism	Multiscale integrated approach. Societal metabolic emphasis. The main mechanism of the model is to check the availability of energy supply to match the requirements of the society for hierarchical levels in each sector. This comparison is subsequently connected to labour. It uses unspecific groups for economic activities and just one scenario is considered.	Increase in jobs if skilled professionals are available: more than 20,000 jobs in the region by 2020. It could compensate for job losses in traditional sectors and, besides, include a portion of the unemployed population.	2000–2050	Rodriguez-Huerta et al. [86]

Source: Own elaboration.

thought to show a higher labour intensity than traditional sources [97].

- The empirically significant variables to explain the results of the transition are: speed of implementation [101], investment [112], imports, exports [15], prices of primary factors, labour flexibility, labour intensity [75], and wage level and elasticity [98].
- While the ILO (2018) [45] states that gender is an important variable for just energy transition analysis, as it could widen inequalities, gender worries are hardly present in the research.

5.2. Distributional effects

The criticism of the German transition is also found in the case of its distributional effects. The main claim is that the financing of the transition and the resulting higher energy prices are regressive, so that the transition and the rise of energy poverty could be linked [101]. This conclusion is, notwithstanding, minimised by other studies [102].

Generally, as Unnerstall [102] noted, most of the studies about Germany are launched from an exclusive point of view, so some authors end up proposing an increasing trust in market mechanisms [113] as a reaction to the failures in the design of Germany's public policies instead of redesigning them to correct the errors, as in any other political process.

As far as the distributional matter is concerned, perceived burdens are of key importance [114], along with the objective effects. The high cost of the transition is detected as a problem by one-fourth of German citizens [115] and many criticise that their distribution is unfair [116]. The preferences of individuals to assume the costs of the transition are correlated with demographic variables. Groh & Ziegler [114] also conclude that the burdens are not as high as perceived, and that the criticism could have been spread by lobbies from the most affected sectors (which do not have to match the most vulnerable sectors). Therefore, this criticism would lack any real basis and, instead, would have been promoted by elites that risk the most in terms of losses caused by the transition.

Usually, the distributional analysis of the German case only focuses on electricity [117–119]. Unnerstall [102] considers that this fixation is linked to the fact that the *Energiewende* is almost exclusively devoted to it. However, electricity is also the protagonist in other contexts [41,78]. This is partly because, in a transition, a major need is the development of electric grids in extension and capacity and the cost of development is imposed on customers. Consequently, areas where further improvements are needed must face a higher burden, for instance in rural areas, where a lower income level is also found. This burden, interlinked with income dynamics, generates greater inequality. The metrics used by Schlesewsky & Winter [117] show that inequality in Germany has been exacerbated by network changes, causing a loss in welfare. Other studies also conclude that even public subsidies to renewables tend to be more beneficial for wealthy citizens [120,121]. This is a frequently detected negative side effect of incentive schemes [65]. Consequently, it is necessary to develop fiscal compensatory measures that balance the situation in terms of distributive impacts, while still supporting the changes.

Frondel et al. [118] specify that prices have doubled in correspondence with the energy transition in Germany, regressively affecting low-income households. Authors compare the rise in German prices with that of Denmark, where electricity tariffs fund the wind deployment, concluding that prices in the Nordic country are higher. In addition, this work connects the rise in prices and its impact on low-income households with higher social conflicts, supporting the need to design compensatory mechanisms.

Gawel et al. [119] offer a different perspective and claim that the discussion on energy prices is not deep enough, so the said effects are not as regressive as they seem. They consider that a part of the socioeconomic problem related to the energy transition is a dramatisation of facts: higher prices can be explained through positive energy quality

increases.

Apart from Germany and Denmark, the increase in energy prices has also been found in other cases, such as the Netherlands. Bulavskaya & Reynès [97] conclude, as described previously, a positive effect of the transition on employment and growth due to the higher capital intensity of renewables. Nevertheless, capital intensity is one of the most prominent elements that negatively cause prices to rise.

In France, the cited EUROGREEN model, analyses environment-energy-income dynamics as a matter on its own, i.e., without relying solely on the impact of rising prices as the rest of studies do. In connection with the findings regarding labour, green growth policies amplify income inequalities, while social equity policies and degrowth notably improves equality at the cost of a higher public deficit (incompatible with the European Union fiscal guidelines) [100].

In contrast to the European cases, a recent study in the United States, focused on wind energy [122], has concluded that there is not a significant distributional injustice regarding income (and ethnicity, note that race has not been explored at a European level), but that there is a relevant unfair impact for younger individuals, with lower levels of qualification and labour force participation, especially in rural areas.

In Table 2, the main empirical studies found to be relevant for this review are classified according to their analytical unit, methodological traits, results and timing:

5.3. Institutional and financial barriers

As this review is focused on labour and income distribution, it is necessary to emphasise that both issues are partly the product of institutional and financial conditions. As found in revisions of historical energy transitions, the evolution of governmental attitudes and their interaction with markets are of a vital relevance to explain the result [23]. Complementarily, in the review of the present transition, institutional frameworks are of vital importance [19] along with the financing, as reflected by the recurrence of significant investment and capital variables in empirical analysis. In contrast, financing tends to be absent in many discussions [124]. That is why this subsection is specifically devoted to gathering the most relevant institutional and financial factors that could block the path to a successful transition.

As cited, the outcome of the transition on labour is determined by such factors as labour mobility [97], skills and educational training, labour market flexibility, labour intensity [75,97,106], wage levels [99] and wage elasticity [98]. These variables are the result of public policies and social institutions, such as collective negotiation and sectorial specificities. A labour policy that does not balance flexibility and social protection, as well as an educational system disconnected from the needs of society, are powerful barriers to reaching a just energy transition.

Other institutional factors can also be concerned. First, the transnational cooperation providing interconnection between grids and the integration of energy markets [95]. Second, the historical and geographical evolution causing structural differences between territories [15,108]. Third, the economic situation and the state of technology determining trade flows [15,74], the production possibility frontier [75], as well as capital intensity [97]. Fourth, the conditions for funding [125,126]. Fifth, the speed of implementation [101], a variable controlled by policymakers, but also determined by demand and infrastructures [105]. A set of policies that do not consider the need to strengthen international cooperation or cooperation between administrations [127], or that obviate the diversities of national territories or the macroeconomic consequences of policy timing, as well as an unfavourable economic scenario, barriers for funding or poor financial conditions, could significantly block the proper development of the transition.

Most importantly, beyond institutional factors, there is an undesirable policy outcome that must be warned against and prevented: the negative side effects of subsidies from a distributional perspective

[65,120,128]. Any transition policy that does not contemplate compensatory fiscal mechanisms could be driven to a massive failure.

Other institutional factors, many times forgotten, are social perceptions, information flows and lobbying practices [114,119], as well as cultural factors [129]. As stated, social perceptions are of key importance, along with the real effects as measured by the objective indicators. These perceptions are a result of information flows and can be influenced by lobbies and elites whose interests are threatened. Distortions and delays in information flows could also be an obstacle.

Finally, from the financial point of view, one of the biggest concerns is related to the financing of the grid development by electricity consumers [117]. Under an adaptive market hypothesis, there would be room to identify more structural barriers, pricing mechanisms, market design influence and behavioural issues in the elaboration of transition policies [124].

6. Research agenda proposal

Based on the previous findings, we have found several methodological gaps, the following being the most notable:

1. An underdevelopment of methodologies applied to the analysis of income distribution. While relevant advances have been made in the field of labour, income impacts and distribution are delayed. Stronger and more developed methodologies are needed.
2. The recurrence of methods that obviate the study of net employment, job quality and working hours. A problematic reuse of factors has also been found.
3. The lack of any consideration of post-growth scenarios in models. While increasing consideration of post-growth has been detected, there is no real translation to empirical analyses.
4. The pre-eminence of the German transition in the literature. Germany cannot be used as the default case study, since negative consequences are recurrently detected in empirical studies.
5. The almost non-existent mention of the precise consequences of the transition for women. A priori, women could be specifically harmed, so inequalities may not only be neglected, but also exacerbated. Not seeing the gender implications of the energy transition is an analytical bias that cannot be afforded any longer.
6. The scarcity of works that aim to analyse the challenge of designing fiscal compensatory mechanisms to correct the negative side effects of incentive schemes, such as the subsidies to promote renewable energies.
7. The methodological insistence on the advantages of IAMs and systemic analytical approaches, such as system dynamics, in contrast to the absence of studies that aspire to apply them jointly.
8. The prevailing use of models that focus exclusively on the energy sector, or that estimate the effect on the whole economy as a consequence of shifts only in the energy sector. These studies also show a narrow perspective, limited to their academic area.
9. The generalised consideration of registered data as the only elements that economic agents use to take decisions, without considering the role that subjective perceptions play as a result of information flows and stakeholders' views.

These gaps evince a significant disconnection between the institutional framework and the methods used by scholars, and between scholars' theoretical considerations and their final empirical applicability.

In order to overcome these limitations, we propose the following research agenda to be developed in future works:

High priority tasks: short to medium term

1. *Increase the efforts put into the research of income levels and distribution.* It must be considered as a matter on its own, rather than a problematic exclusively linked to the consequences of labour shifts

or the rise in energy prices.

2. *Widen the variety of methods applied in the case of labour.* Research should aim to capture the big picture and focus on the net employment balance and widen the knowledge of job quality and working time variations. The reuse of employment factors should be prevented.
3. *Introduce the possibility of controlled degrowth in models, away from the predominant growth paradigms.* Both growth and degrowth should be considered to test the models and the political outcomes. Otherwise, modelling is biased towards growth preferences.
4. *Enlarge the number and type of geographic units proposed.* This is why, after revising the available studies and data, we propose the Nordic countries as a possible case instead, especially Sweden and Denmark.
5. *Start considering seriously the gender dimension in the energy transition, following the recommendations from the ILO.* In the process of calculation, a disaggregation of workers should be done according to their gender, so negative impacts on women's labour can be detected, as well as testing the effects of equality policies on a transition scenario (for instance, family reconciliation with respect to the variation of working time).

Medium priority tasks: medium to long term

1. *Explore the possibilities for modelling techniques that hybrid IAMs and system dynamics offer, especially when applied jointly.* There are still many methodological borders separating them; the categorical classification discussed being a proof. As long as a joint application of methods serves to reinforce the strength and consistency of the analysis, it is an interesting exercise to progressively blur such borders.
2. *Broaden the scope of models and estimate the effects of the energy transition in the economy, developing a more solid macroeconomic base, directly sector by sector, rather than solely focusing on the energy sector or studying the diffusion from the energy sector to the rest of the economy.* Models with diverse thematic modules relying on trans-disciplinary fields could be a methodological possibility to explore. Besides, there is a need to deepen our knowledge on fiscal compensatory mechanisms.
3. *Developing techniques that allow researchers to consider citizens' perceptions along with real data in models.* Even though research must be built on verifiable available data, citizens, who are the principal agents and the main recipients of costs and benefits, do not consider this kind of data on a continuous basis, but rely more on perceptions. Perceptions are not as easily suited to modelling as real data. However, a growing consideration should be given to this aspect, with the goal of understanding the ultimate underlying reality on a social level.

7. Conclusions

The global environmental and human crisis has led to the renaissance of just transitions, this time centred around the shift to a sustainable energy mix. This renewed notion of just energy transition, which has transcended its origin in the US union movement of the past century, has been sponsored in the framework of the Sustainable Development Goals by the International Labour Organisation. The ILO has stressed both sides of the transition: the one where the transition creates net employment, improvements in the quality of jobs and a reduction of poverty; and the dark side, where the transition implies costly socioeconomic restructuring and regressive political effects, while facing the bill of climate change adaptation.

This systematic review has revealed some key insights into the methodological approaches in the field.

First, the diversity of methods has motivated some restrictive classifications in previous studies that must be observed with caution and an open perspective, since these classifications have been surpassed by

the empirical state-of-the-art. Empirical analyses are blurring the lines between methods and generating methodological combinations. Notwithstanding, there is a notable preference for Computable General Equilibrium models and Integrated Assessment Models based on Input-Output tables. This review has detected a growing interest in hybrid approaches, that combine top-down and bottom-up modelling, as well as interest in systemic modelling techniques such as system dynamics. There is also a pre-eminence of case studies focused on the research of impacts at a national and regional level, many times with a comparative ambition. Lately, regional levels are gaining momentum to determine asymmetric consequences among territories.

Second, there is a scarcity of works devoted to the distributional consequences of the transition and an underdevelopment of methodologies in comparison with the abundant literature and advanced methodology found in the case of labour. Besides the lack of synchronisation between the two topics, both have been primarily analysed through a linear and unidirectional perspective; while in fact, the transition is a complex, multidimensional and multidirectional phenomenon. Likewise, many studies have put their efforts into estimating direct gross impacts on employment levels, when the focus to assess the just transition should be on direct, indirect and induced net volumes. Even if this misconception of employment balances has been increasingly prevented, most studies are blind to impacts beyond volumes, such as the quality of jobs, the variations in working hours and the gender balance of the transition. These absences evince the disconnection between the institutional approach, reflected by the ILO, and the academic approach, and also between scholars' theoretical considerations and their final empirical applications.

Third, the key variables identified in the literature to determine the outcome of the transition are the timing/speed of policy implementation, investment, international trade, the production possibilities frontier, the prices of primary factors, labour flexibility and intensity, and the elasticity of wage levels. Studies generally find a positive, yet relatively small, impact on employment levels, led by the construction and energy sectors, in conjunction with a negative distributional effect caused by rising electricity prices as a result of network developments and the regressive side effects of public subsidies. Neoclassical models have reached less optimistic results than other methodological backgrounds and tend to emphasise the role of investment and wage levels, while relegating skill shortages. Methodologies relying on surveys and workshops highlight the negative distributional effects with more intensity. This result points to the necessity to consider perceptions alongside real data in the assessment of the transition.

In the light of these conclusions, we propose a research agenda that underlines the need to make greater efforts in the study of income dynamics, the effects on job quality and working hours and the role of supporting such policies as compensatory fiscal mechanisms, specifically considering the precise impact of the transition pathways on women. There is also a need to introduce post-growth scenarios in the models to complement green growth strategies and to broaden the geographic scope being researched: instead of Germany, that monopolises the available literature, it would be enriching to consider the Nordic countries, especially Sweden and Denmark, due to their institutional backgrounds.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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